

Computing the Posterior Probability of Optimality for Factor-Gaussian Beliefs

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Abstract

Thompson sampling plays each alternative with the posterior probability that it is best. Methods that need that probability itself approximate it from marginal means and variances. We compute it directly, for all n alternatives at once in time linear in n , whenever the posterior covariance is a low-rank factor term plus a diagonal. Simulation under common random numbers and a Gaussian bandit with a factor prior both produce such posteriors exactly. In both, the computed vector stops sooner than the marginal approximations at the same nominal threshold, and incurs less regret.

1 Introduction

An analyst has twenty candidate inventory policies and estimates each one’s expected profit by simulation. A website has fifty advertisements and records clicks on whichever one it shows. A chemist has predicted binding strengths for a thousand candidate molecules. In each case there are n alternatives, and alternative i has an unknown value θ_i : the policy’s true expected profit, the advertisement’s true click rate, the molecule’s true binding strength. The data $\mathcal{D}$ collected so far give a posterior belief about $\theta = (\theta_1, \dots, \theta_n)$.

The posterior probability of optimality is the vector

$$p_i = \Pr\left(\theta_i = \max_j \theta_j \mid \mathcal{D}\right), \quad i = 1, \dots, n. \quad (1)$$

Its entry p_i is the probability, given the data, that alternative i is truly the best. If $p_3 = 0.9$ for the inventory policies, there is a ninety percent chance that policy 3 has the highest expected profit.

Decisions about where to collect data use this vector. The analyst can stop simulating once the largest p_i exceeds 0.95, and until then can spend runs on the policies whose p_i are close. The website can show advertisement i with probability p_i , which is Thompson sampling. A rule that must respect constraints, or must be reproducible without randomness, needs the numbers p_i themselves.

When the beliefs about different alternatives are independent, each p_i is a one-dimensional integral. In practice they are usually correlated, because alternatives share causes. Inventory policies simulated on the same demand history all do well on a quiet day and badly on a busy one. Advertisements aimed at the same audience have similar click rates. Molecules built on the same scaffold bind similarly. With correlated beliefs, p_i is an n -dimensional Gaussian orthant probability, and no general fast method computes it.

Two literatures need this vector and approximate it. The first is ranking and selection, which chooses the best of n simulated systems. Its procedures bound the probability of a correct selection with Bonferroni-type inequalities, or estimate it by posterior sampling. The joint vector is not evaluated.

The second is Bayesian bandits and reinforcement learning. [Tarbouriech et al. \(2023\)](#) call Equation (1) the key object of that field and write that computing it “involves computing several complicated integrals with respect to the posterior and is intractable in most cases.” Their method, VAPOR, replaces it with a variational program. [Menet et al. \(2026\)](#) fine-tune a large language model toward a policy that uses only the marginal means and standard deviations of the posterior. Section 4 describes these approximations.

Correlation that comes from a few shared causes has a simple form. Each value is its own part plus a weighted sum of ρ shared variables:

$$\theta_i = m_i + w_i' z + \varepsilon_i, \quad z \sim N(0, I_\rho), \quad \varepsilon_i \sim N(0, d_i), \quad (2)$$

with all the ε_i independent of each other and of z . For the inventory policies, z might be the overall demand level and its volatility. The row w_i says how strongly policy i responds to each. The covariance of θ is then

$$\Sigma = WW' + \text{diag}(d), \quad W \in \mathbb{R}^{n \times \rho}, \quad \rho \ll n, \quad (3)$$

where W has rows w_i' . The variables in z are called factors, and W holds the loadings.

For beliefs of this form, Equation (1) can be computed for all n alternatives at once, in time linear in n . Section 2 gives the computation, which is the pricing pass for correlated contests in [Cotton \(2026\)](#). Section 3 shows that one standard model from each literature produces a posterior of exactly this form after every update. Section 4 relates the vector to Thompson sampling and to the approximations above. Sections 5 and 6 measure what using the vector changes, in a simulation stopping problem and in a bandit. Section 7 states where the exact form fails.

2 Computing the vector for beliefs with shared factors

Write the posterior in the form of Equation (2), $\theta = m + Wz + \varepsilon$. Assume every $d_i > 0$, so the maximum is almost surely unique and the p_i sum to one.

The shared factors z are what make the alternatives dependent. Fix z , and nothing is shared: the alternatives are independent, with means $m_i + w_i' z$ and variances d_i . The probability that i is best given z is then a one-dimensional integral, and p_i is its average over z .

Let $F_{i|z}$ and $f_{i|z}$ be the conditional distribution and density functions of θ_i given z . Alternative i is the maximum when every other alternative falls below it, so

$$p_i = \int_{\mathbb{R}^\rho} \phi_\rho(z) \int_{\mathbb{R}} f_{i|z}(x) \prod_{j \neq i} F_{j|z}(x) dx dz. \quad (4)$$

The inner integrands for different i share the product $G_z(x) = \prod_j F_{j|z}(x)$. Dividing out the i th term gives every p_i from the same product:

$$p_i = \mathbb{E}_z \int f_{i|z}(x) \frac{G_z(x)}{F_{i|z}(x)} dx. \quad (5)$$

On a lattice of L points in x , with Q quadrature nodes in z , the whole vector costs $O(nLQ)$. The accuracy depends on L , on the lattice window and on Q , and the Q needed grows with the number of factors and the geometry of W .

The integral is exact, but its evaluation is numerical. With three alternatives, $W = (0, 1, 3)'$ and $d_i = 0.1$, the vector is known analytically from the bivariate quadrant identity, and the default settings of the implementation are in error by total variation 0.0015. Raising the factor quadrature resolution reduces this error.

The numerical details are in [Cotton \(2026\)](#): lattice placement, products taken in logs, and the tails. That implementation computes the probability of being the minimum, and the maximum is obtained by negating m and W . It is used here without other change. The computed vector was compared with argmax frequencies from 2×10^5 Monte Carlo draws at sixteen posteriors: up to four per regime in the stopping experiment, at the twentieth replicate, and two per regime in the bandit, at round 50. The largest total-variation distance in those comparisons is 0.0042, which is at the Monte Carlo noise level for that sample size. These comparisons are statistical checks, not error bounds.

3 Two models whose posteriors have exact factor form

The computation of Section 2 needs the posterior covariance in the form of Equation (3). Two standard models produce it exactly after every update, one from each literature. In simulation, the shared factors come from running every system on the same random inputs. In the bandit, they come from a prior in which arms share features.

3.1 Simulation with common random numbers

A simulation study compares n systems, such as the inventory policies of the introduction, by running each one many times and averaging its output. Each run is called a replicate. The unknown θ_i is system i 's long-run average output.

A standard way to compare systems more precisely is to give every system the same random inputs in each replicate. This is called common random numbers. In replicate r , all twenty inventory policies face the same simulated demand history. A high-demand history raises every policy's profit, so the outputs move together, and the differences between policies are measured more precisely than their levels.

Summarise the shared random inputs of replicate r by ρ variables F_r , such as the demand level and its volatility. System i responds to them with loadings v_i and has its own noise ε_{ir} :

$$Y_{ir} = \theta_i + v_i' F_r + \varepsilon_{ir}, \quad F_r \sim N(0, I_\rho), \quad \varepsilon_{ir} \sim N(0, \sigma_i^2). \quad (6)$$

The F_r and ε_{ir} are independent of each other and across replicates. The loadings v_i and noise scales $\sigma_i > 0$ are known properties of the simulation. The unknown is θ , with prior $\theta \sim N(0, s_0^2 I)$ and $s_0 > 0$.

The posterior covariance of θ turns out to have factor form, with the same number of factors ρ . The derivation takes four steps.

The covariance of one replicate across systems is

$$\Lambda = VV' + D, \quad D = \text{diag}(\sigma_1^2, \dots, \sigma_n^2). \quad (7)$$

After $n_r \geq 1$ replicates with mean vector $\bar{Y}$, the posterior of θ is Gaussian with precision

$$A = s_0^{-2}I + n_r\Lambda^{-1}. \quad (8)$$

The first step inverts Λ with the Woodbury identity:

$$\Lambda^{-1} = D^{-1} - D^{-1}VM^{-1}V'D^{-1}, \quad M = I_\rho + V'D^{-1}V. \quad (9)$$

With $U = D^{-1}V$ and $S = n_rM^{-1}$, the precision becomes

$$A = \text{diag}(a) - USU', \quad a_i = s_0^{-2} + n_r\sigma_i^{-2}. \quad (10)$$

This is a positive diagonal minus a positive semidefinite term of rank at most ρ .

The second step inverts A , again with the Woodbury identity:

$$A^{-1} = \text{diag}(a)^{-1} + \text{diag}(a)^{-1}UCU'\text{diag}(a)^{-1}, \quad C = (S^{-1} - U'\text{diag}(a)^{-1}U)^{-1}. \quad (11)$$

The third step shows that C is positive definite, so that it has a square root. Let $K = \begin{pmatrix} \text{diag}(a) & U \\ U' & S^{-1} \end{pmatrix}$. The Schur complement of S^{-1} in K is $\text{diag}(a) - USU' = A$, which is positive definite, and S^{-1} is positive definite, so K is positive definite. The Schur complement of $\text{diag}(a)$ in K is C^{-1} , so C^{-1} is positive definite too.

The fourth step reads off the factor form:

$$\Sigma_{\text{post}} = \text{diag}(d) + WW', \quad d_i = a_i^{-1}, \quad W = \text{diag}(a)^{-1}UC^{1/2}, \quad (12)$$

exactly, for every $n_r \geq 1$. Before any replicate the posterior is the diagonal prior. The posterior mean is $m = A^{-1}h$ with $h = n_r\Lambda^{-1}\bar{Y}$, and both factors are available from Equations (9) and (11).

No step approximates. The posterior has factor form because the shared random inputs have factor form. The number of posterior factors is at most ρ , the number of shared input variables, however many systems are compared.

3.2 A bandit whose arms share features

In a bandit, a decision maker repeatedly picks one of n arms and observes a noisy reward from it. Here θ_i is arm i 's mean reward, such as an advertisement's click rate. Pulling arm a returns $\theta_a + N(0, \sigma_n^2)$, with $\sigma_n > 0$.

The correlation comes from the prior. Advertisements that share an audience or a design are expected to have similar click rates. A linear model expresses this: arm i has known features v_i , and $\theta_i = v_i'\beta + e_i$, with a shared $\beta \sim N(0, I_\rho)$ and independent $e_i \sim N(0, d_{0i})$, $d_{0i} > 0$. The prior is then $\theta \sim N(0, \Sigma_0)$ with $\Sigma_0 = VV' + \text{diag}(d_0)$, which has factor form.

Gaussian-process priors have this form only when the approximation replaces the conditional residual by a diagonal. A standard inducing-point posterior has a dense residual (Titsias, 2009), so for those the diagonal is an additional approximation.

With per-arm pull counts c , the posterior precision is

$$A = \Sigma_0^{-1} + \sigma_n^{-2} \text{diag}(c). \quad (13)$$

Equation (9) applied to Σ_0 puts A in the form of Equation (10), with $a_i = d_{0i}^{-1} + c_i\sigma_n^{-2}$, $S = M_0^{-1}$ and $M_0 = I_\rho + V'\text{diag}(d_0)^{-1}V$. The remaining steps of Section 3.1 apply unchanged. The posterior is exactly factor plus diagonal after every pull, with no more factors than the prior.

4 Thompson sampling and the approximations of the vector

Bandit methods relate to the vector in two ways. Thompson sampling uses it implicitly, without computing it. Two other methods compute approximations of it from marginal means and variances. The experiments of Sections 5 and 6 compare all of them with the computed vector.

4.1 Thompson sampling plays the vector

Thompson sampling draws one sample $\tilde{\theta}$ from the posterior and pulls the arm with the largest sampled value. The probability that it pulls arm i is the probability that arm i is largest under the posterior, which is p_i . Lemma 8 of Tarbouriech et al. (2023) states the general version: the expected occupancy of Thompson sampling equals the posterior probability of optimality.

Pulling an arm with probability p_i from the computed vector, called *exact matching* below, is therefore Thompson sampling in distribution, up to the quadrature error of Section 2. The two have the same expected regret, and Section 6 compares them.

The computed vector gives what a single draw cannot: the numbers themselves. A stopping rule needs $\max_i p_i$, an allocator needs to know which p_i are close, and a reproducible policy needs the probabilities it would otherwise sample.

4.2 Approximations from marginal means and variances

When the numbers p_i are needed, current methods approximate them using only each alternative’s marginal mean m_i and marginal standard deviation $s_i = \sqrt{d_i + \|w_i\|^2}$. Both ignore the correlation between alternatives.

F-LITE (Menet et al., 2025) treats the alternatives as independent, $\prod_i N(m_i, s_i^2)$, and approximates the probability of being best by $\Phi((m_i - \kappa^*)/s_i)$, with a threshold κ^* chosen so that these sum to one. The threshold step is a second approximation. With two independent arms, $N(1, 1)$ and $N(0, 1)$, it gives 0.691 for the first arm, against the exact 0.760. F-LITE therefore departs from p even when the posterior is independent.

The *VBOS policy* is the variational Bayesian optimistic sampling rule of O’Donoghue and Lattimore (2021), extended by Tarbouriech et al. (2023). In the near-closed form of Menet et al. (2026) it is

$$\tilde{\pi}_i = v\left(\frac{m_i - \kappa^*}{s_i}\right), \quad v(c) = \exp\left(-\frac{1}{8}(\sqrt{c^2 + 4} - c)^2\right), \quad (14)$$

with κ^* again chosen so the vector sums to one.

The VBOS policy departs from p on purpose, and its regret analysis depends on that departure. It is a policy, not an estimate of p . Its distance from p is reported below because Menet et al. (2026) use it as the trainable substitute for p .

5 Stopping under common random numbers

The experiment uses the posterior of Section 3.1 with $n = 100$ systems, $\rho = 2$ scenario streams and prior scale $s_0 = 1$. True means are drawn from the prior. Every system has replicate variance one, so each replicate carries the same marginal information in every regime. The *factor share* is the fraction of that variance that comes from the shared scenario.

The regimes differ only in how systems load on the scenario streams:

- *Independent*: factor share zero. This is the control.
- *Aligned*: every system loads on the same stream, with factor share 0.6. Differences between systems then vary less than their marginal variances suggest.
- *Opposed*: half the systems load positively and half negatively on one stream, with factor share 0.6. Differences between systems in opposite halves then vary more than their marginal variances suggest.

Three rules compete. Each computes its own estimate of p after every replicate, stops the first time its largest entry reaches $1 - \delta$ with $\delta = 0.05$, and selects that entry. The exact rule uses the computation of Section 2. The other two use F-LITE and the VBOS policy.

For the true vector the rule has a Bayesian guarantee. Let τ be the first time $\max_i p_i \geq 1 - \delta$, let B be the budget, and on $\{\tau \leq B\}$ let I be the selected alternative and I^* the best. Then

$$\Pr(I \neq I^* \mid \tau \leq B) = \mathbb{E}[1 - p_I(\mathcal{D}_\tau) \mid \tau \leq B] \leq \delta. \quad (15)$$

Optional stopping does not affect this statement, which averages over the prior. It is not a frequentist guarantee for a fixed θ , and it applies to the true p , not to a numerical approximation of it.

The three rules share a nominal threshold, not a true confidence level. With two independent arms of equal posterior standard deviation s , an entry of 0.95 requires a mean gap of 2.33 s for the exact vector, 3.29 s for F-LITE and 4.84 s for the VBOS policy. The true posterior probability of a correct selection at those gaps is 0.950, 0.990 and 0.9997. Part of any extra sampling by the marginal rules is this higher effective target.

regime	rule	mean replicates	realized PCS	censored of 200
independent	exact	141.2	0.970	35
	F-LITE	207.0	0.985	65
	VBOS	271.5	1.000	102
aligned	exact	69.7	0.958	9
	F-LITE	175.6	1.000	60
	VBOS	240.1	1.000	91
opposed	exact	106.3	0.944	21
	F-LITE	171.7	0.979	57
	VBOS	239.0	0.991	88

Table 1: Stopping at nominal 95% confidence. Each regime has 200 replications and a budget of 400 replicates. Mean replicates is the mean of the stopping time capped at 400, so a censored run counts 400. Realized PCS is the fraction of stopped runs that selected the true best system. The stopped runs differ by rule.

In Table 1, every rule’s realized PCS among its stopped runs is at or above nominal, within binomial noise. The exact rule’s 0.944 in the opposed regime has 179 stopped runs and a standard error of 0.016.

At this common nominal threshold and budget, the two marginal-only rules used 1.5 to 3.4 times the capped mean replicates of the exact rule, and hit the budget far more often. The gap

is largest in the aligned regime, where the leading systems’ differences are tightly coupled. There the exact rule stops after 69.7 replicates on average, against 175.6 and 240.1, and stops in 191 of 200 runs, against 140 and 109. An efficiency comparison at a matched true error would need thresholds calibrated to a common PCS, which this experiment does not do.

The marginal-only estimates are far from the exact vector along the way. Before stopping in the aligned regime, F-LITE is at median total-variation distance 0.123 from the exact vector, with ninetieth percentile 0.174. The VBOS policy is at 0.141, with ninetieth percentile 0.206.

The distances do not vanish in the independent regime, where they are 0.063 and 0.096. F-LITE’s threshold step is biased even when the posterior is independent, and the VBOS policy departs from p by design. At the stopping experiment’s Monte Carlo checks, the largest total-variation distance of the computed vector was 0.0015.

6 A correlated Gaussian bandit

The bandit uses the posterior of Section 3.2 with $n = 50$ arms, prior variance one per arm, observation noise $\sigma_n = 1$, horizon 200 and 100 replications. The prior’s factor term has two columns. In the *independent* regime it is zero. In the *clusters* regime each half of the arms loads on its own factor, which carries 0.75 of the prior variance.

Four policies act on the same posterior: Thompson sampling, exact matching, sampling from the VBOS policy, and sampling from F-LITE. Table 2 reports cumulative Bayes regret and, at the horizon, whether the arm with the highest posterior mean is the true best and the shortfall of that arm’s value below the best. Its last column is the total-variation distance between each policy’s action distribution and the exact vector at mid-horizon, along that policy’s own trajectory.

regime	policy	regret	identification	shortfall	TV to exact p
independent	Thompson	172.1	0.72	0.073	0
	exact matching	173.9	0.75	0.055	0
	VBOS	190.3	0.76	0.061	0.159
	F-LITE	179.2	0.79	0.052	0.057
clusters	Thompson	122.7	0.61	0.097	0
	exact matching	124.8	0.51	0.113	0
	VBOS	135.9	0.59	0.105	0.129
	F-LITE	128.7	0.51	0.121	0.061

Table 2: Mean over 100 replications of cumulative Bayes regret, the rate at which the highest posterior mean identifies the true best arm at the horizon, and the shortfall of that arm. The last column is the total-variation distance from the computed vector at mid-horizon. It is zero for exact matching by construction. For Thompson sampling it is zero relative to the true p by Lemma 8, and is entered as zero rather than measured.

Thompson sampling and exact matching have the same expected regret by Lemma 8, and their sample means differ by one to two percent of cumulative regret in both regimes. The VBOS policy’s mean regret is ten to eleven percent above Thompson sampling’s, and its action distribution is at total-variation distance 0.13 to 0.16 from the computed vector. F-LITE’s is

four to five percent above. Per-replication regrets were not saved, so these differences carry no paired intervals.

The identification rates do not separate the policies. Their differences are within two binomial standard errors at 100 replications. The regret and distance columns carry the comparison.

7 Scope

The derivations of Section 3 are exact because the observations there are generated with factor structure. When scenario effects enter a simulation nonlinearly, a low-rank model of Λ is an approximation, and the posterior inherits its error. When a Gaussian-process kernel is dense with a short length scale, a factor fit has to precede the computation, and its residual is the cost.

The probability of optimality is not the knowledge gradient. Under simple-regret loss, the one-step value of a measurement is the knowledge-gradient quantity (Frazier et al., 2008), an expectation over the next posterior means. Tuisov (2026) shows that this is the normative myopic value of computation, and notes that the exact dynamic version “is itself generally intractable.” Neither quantity is computed here.

The regret analysis of O’Donoghue and Lattimore (2021) depends on the VBOS tilt. Its measured distance from p quantifies the approximation, but does not by itself make a case against the tilt.

Thompson sampling needs one posterior sample and no vector. The exact vector matters where a sample is not enough: stopping, allocation, constraints and derandomization. Both experiments above are of that kind.

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